



## **Assignment 01**

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| <b>Roll No</b>      | <b>FL-21565</b>           |
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# 1. Requirement Specification Document:

## ➤ Functional Requirements

### 1. User Management

#### 1.1 Add New User (Librarian/Student/Member)

- **Purpose:** Register new users (librarians, students, or members) into the system.
- **Details:**
  - Collect details like **name, ID, contact info, and role**.
  - Assign a unique **User ID** for identification.
  - Store user data in a database.

#### 1.2 Edit/Update User Information

- **Purpose:** Modify existing user details (e.g., phone number, address).
- **Details:**
  - Search for a user by **ID or name**.
  - Allow updates to personal information.

#### 1.3 Delete User

- **Purpose:** Remove a user from the system (e.g., if they leave the institution).
- **Details:**
  - Search for the user and confirm deletion.
  - Ensure no active book loans before deletion.

#### 1.4 View User Details

- **Purpose:** Display user information for administrative purposes.
- **Details:**
  - Search by **ID, name, or role**.
  - Show borrowing history if applicable.

## 2. Book Management

### 2.1 Add New Book

- **Purpose:** Enter a new book into the library database.
- **Details:**
  - Input **title, author, ISBN, genre, publication year, and quantity**.
  - Assign a unique **Book ID**.

### 2.2 Update Book Information

- **Purpose:** Modify book details (e.g., change quantity, correct author name).
- **Details:**
  - Search by **title, ISBN, or Book ID**.
  - Allow editing of book metadata.

### 2.3 Remove Book

- **Purpose:** Delete a book from the system (e.g., if lost or damaged).
- **Details:**
  - Ensure no active loans before removal.
  - Log deletion for records.

### 2.4 Search Books

- **Purpose:** Allow users to find books easily.
- **Details:**
  - Search by **title, author, ISBN, or genre**.
  - Display availability status (**Available/Borrowed**).

## 3. Borrowing & Returning System

### 3.1 Issue Book (Check-out)

- **Purpose:** Allow users to borrow books.
- **Details:**
  - Verify user eligibility (no overdue books).
  - Record **check-out date** and **due date**.
  - Update book status to "**Borrowed**".

### 3.2 Return Book

- **Purpose:** Process book returns.
- **Details:**
  - Scan **Book ID** or enter manually.
  - Check for **late returns** and apply fines if needed.
  - Update book status to "**Available**".

### 3.3 Renew Book

- **Purpose:** Extend borrowing period if no reservations exist.
- **Details:**
  - Check if renewal is allowed (max renewals = 1 or 2).
  - Update **due date**.

### 3.4 Track Overdue Books

- **Purpose:** Identify late returns and manage penalties.
- **Details:**
  - Automatically flag overdue books.
  - Calculate fines (**e.g., \$1 per day late**).
  - Send reminders via email (if implemented).

## 4. Reporting & Analytics

### 4.1 Generate Borrowing Reports

- **Purpose:** Track book circulation trends.
- **Details:**
  - Show **most borrowed books**.
  - Filter by **date range, genre, or user type**.

### 4.2 View Fines & Payments

- **Purpose:** Manage late fees and payments.
- **Details:**
  - List users with pending fines.
  - Record payments and update balances.

### 4.3 Inventory Status

- **Purpose:** Check total books, available copies, and lost/damaged books.
- **Details:**
  - Display summary statistics.
  - Export data to **CSV/PDF** (optional).

## 5. Authentication & Security

### 5.1 Login/Logout

- **Purpose:** Secure access to the system.
- **Details:**
  - **Librarians** have full access.
  - **Members** can only view/borrow books.

### 5.2 Password Reset

- **Purpose:** Allow users to recover accounts.
- **Details:**
  - Email verification or security questions.

## ➤ **Non-Functional Requirements**

### **1. Performance:**

- System should handle **100+ concurrent users**.
- Search results should load in **<2 seconds**.

### **2. Security:**

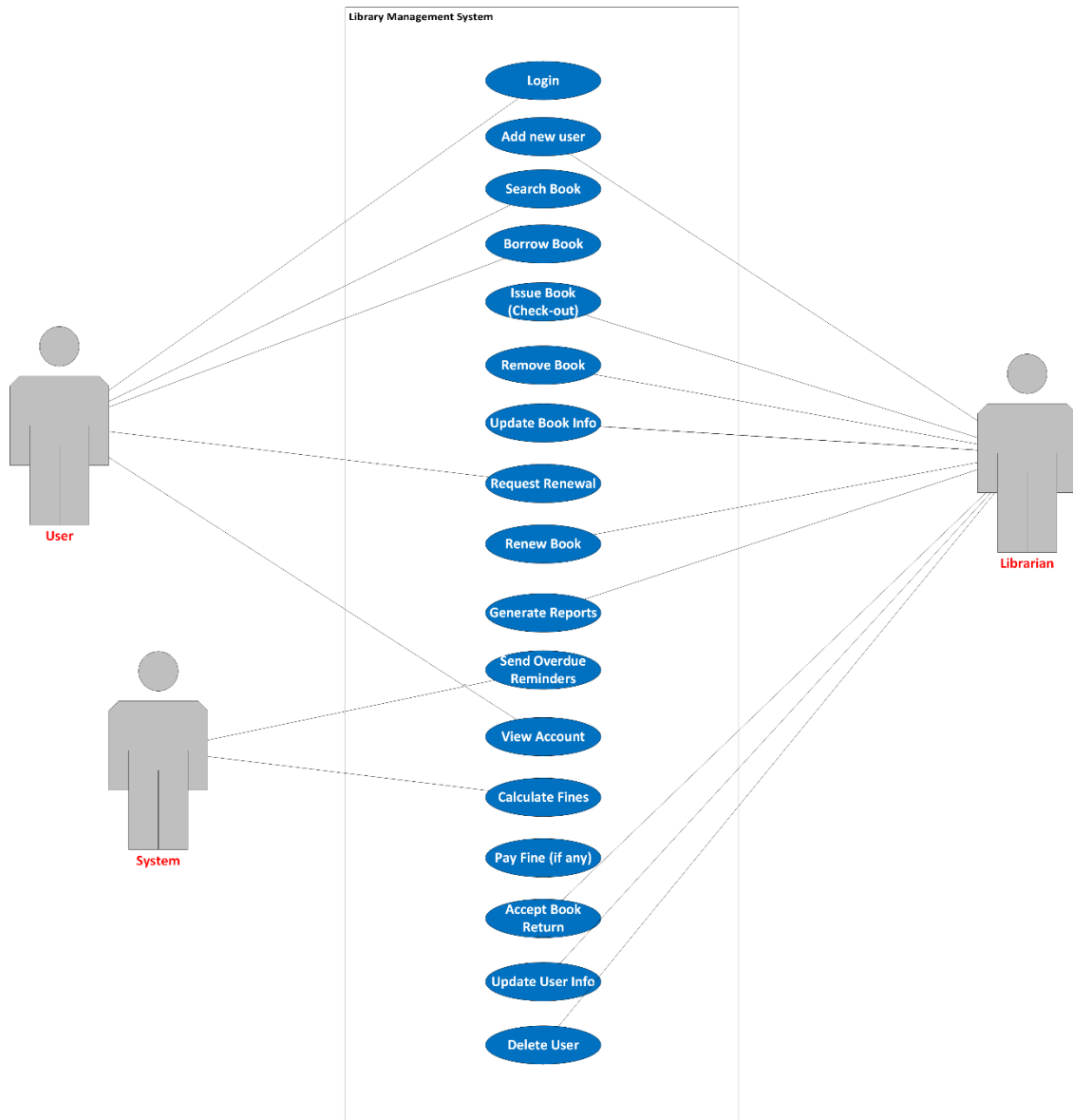
- Passwords must be **encrypted**.
- Unauthorized users **cannot modify data**.

### **3. Usability:**

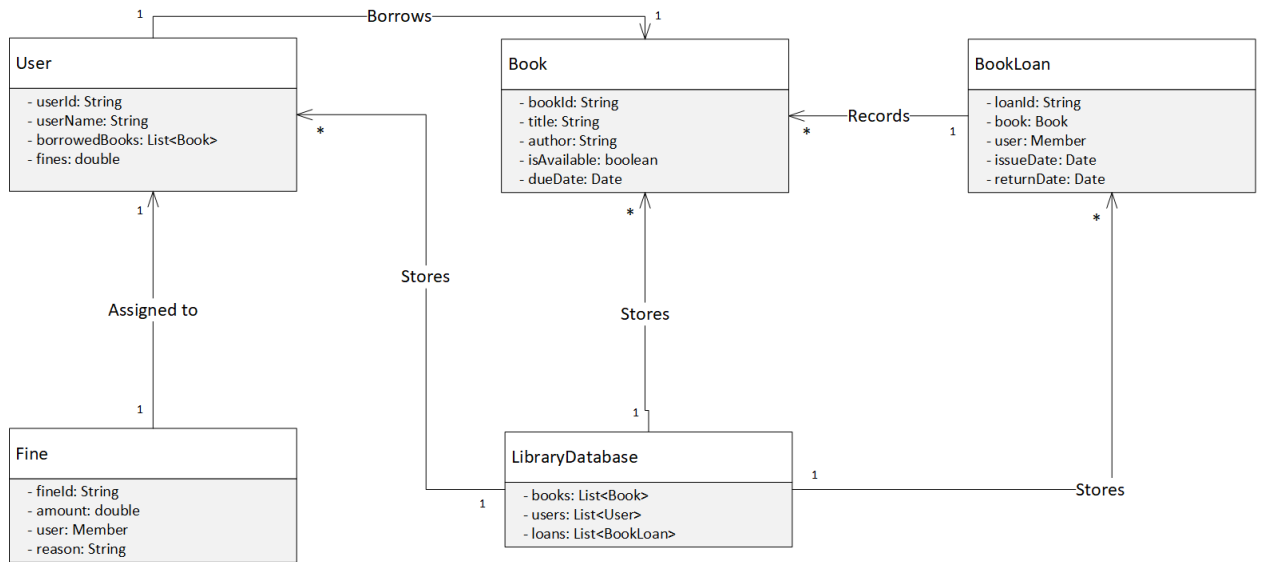
- Simple, intuitive interface.
- Mobile-friendly (optional).

## 2. Detailed Design Document:

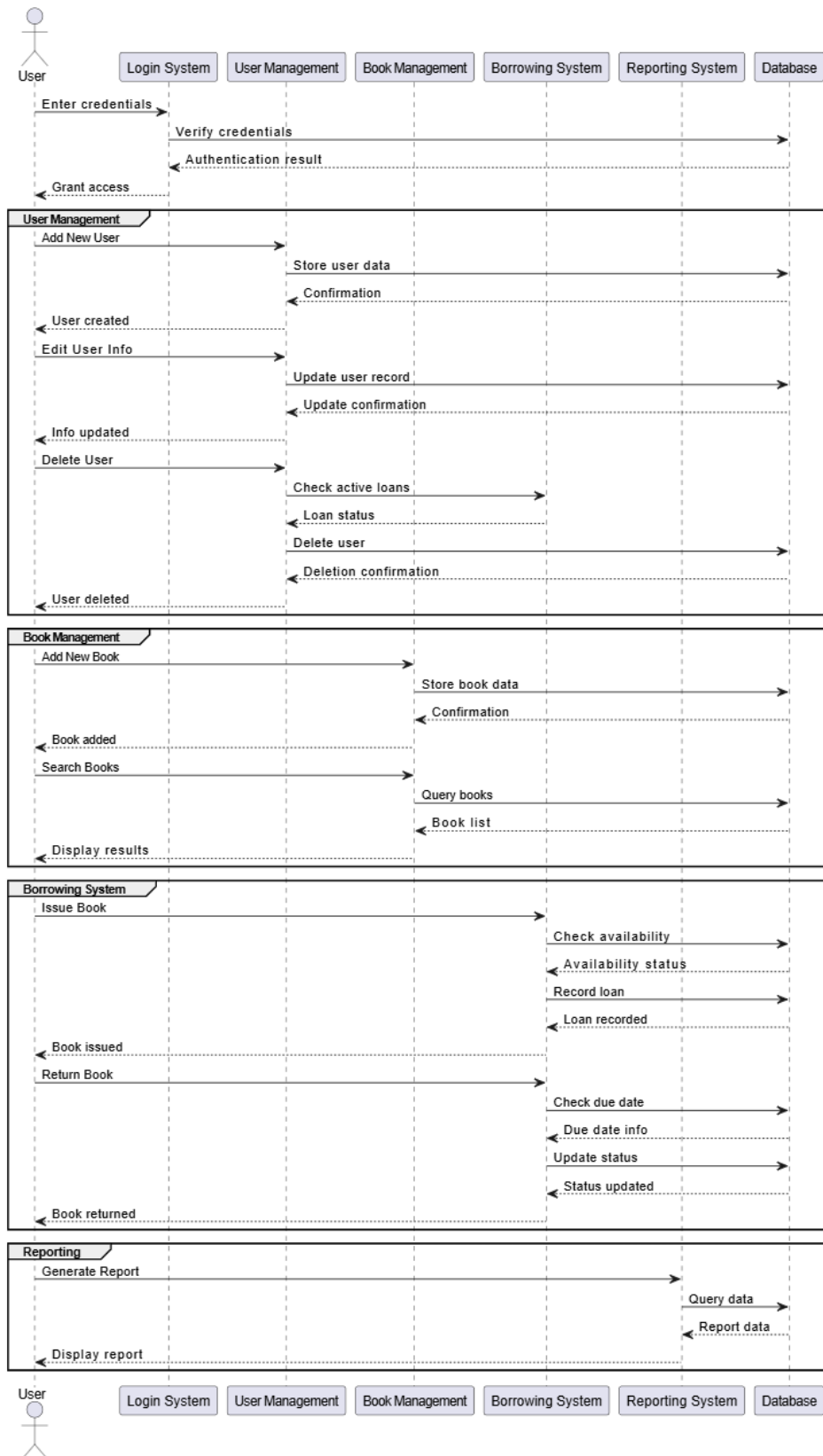
- Use-case diagram:



- Object diagram:



## • Sequence diagram:



- **Communication diagram:**

